Predicting NFL Margin of Victory: *Machine Learning vs. Vegas*

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Research Question

- A foundational sports gambling concept is the **points spread**: handicap assigned to one of two teams to theoretically give each team an equal chance at winning in the eyes of the bettor
  - i.e. if Team A is a clear favorite over Team B, then oddsmakers make set a -5.5 point spread where Team A has to win by 6 points or more for a points spread bet on Team A to be victorious.

- Is it possible to create a machine learning framework to predict margin of victory for any given NFL matchup, specifically with the goal of a win rate that is at least 52.4%?
  - Success potential comes from prediction being supplied solely off of data and past outcomes whereas the point spread is not necessarily an unbiased predictor of the game outcome

Data

- Training dataset consists of offensive + defense statistics for >1000 games from 2000-2007 NFL games (1544 total)
- Test dataset is comprised of 2008-2009 NFL games (390 total)
- For each game, the features included are (all of which have data for the home team, away team, home team streak, and away team streak):
  - Winning %
  - Points Per Game Scored, Points Per Game Allowed
  - Total Yards Per Game Gained, Total Yards Per Game Allowed
  - Pass Yards Per Game Gained, Pass Yards Per Game Allowed
  - Rush Yards Per Game Gained, Rush Yards Per Game Allowed
  - Turnovers taken per game, Turnovers given up per game
- Two additional features are computed strength (home team / away team) and temperature difference

Data (cont.)

- Goal of temperature difference is to quantify the impact of home field advantage in the model
- Computed strength is a weighted average of the strengths of its opponent

$$s_i = \frac{1}{n_i}\sum_{j=1}^{N} a_{ij} r_j \longrightarrow A\vec{r} = \lambda\vec{r}$$

- where $\mathbf{a_{ij}}$ is a nonnegative number depending on the outcome of the game between team *i* and team *j*

$$a_{ij} = h\left(\frac{S_{ij}+1}{S_{ij}+S_{ji}+2}\right) \qquad h(x) = \frac{1}{2} + \frac{1}{2} sgn(x - \frac{1}{2})\sqrt{|2x-1|}$$

- **h(x)** is designed such that it is important to win a given matchup but not as important to run up the score

Modeling

- **Gaussian Process (GP) Model**
  - Probability distribution over functions → collection of random variables such that any finite set of them has a joint multivariate normal distribution
  - It is fully specified by a mean, **m**, and covariance function, **k**:

$$m(\vec{x}) = \mathbb{E}[f(\vec{x})]$$
$$k(\vec{x}, \vec{x}') = \mathbb{E}[(f(\vec{x}) - m(\vec{x}))(f(\vec{x}') - m(\vec{x}'))]$$

- We can write the GP as: $\boxed{f(\vec{x}) \sim GP(m(\vec{x}), k(\vec{x}, \vec{x}'))}$
  - where we consider zero mean and a squared exponential covariance

- Margin of victory is modeled as y = f(x)+ϵ where $\epsilon \sim N(0, \sigma_n^2)$ is i.i.d. Using the GP prior, the joint distribution of training outputs and test predictions is multivariate Gaussian, allowing closed form posterior mean and variance for predictions

- The posterior mean, **f***, gives the predicted spread and the posterior covariance gives the uncertainty (diagonal entries of **K** provide variance for each prediction):

$$\bar{f}^* = K(X^*, X)[K(X, X) + \sigma_n^2 I]^{-1} y$$
$$\bar{K} = K(X^*, X^*) - K(X^*, X)[K(X, X) + \sigma_n^2 I]^{-1} K(X, X^*)$$

(Kernel hyperparameters optimized by maximizing the log likelihood corresponding to this model)

Feature Selection

- **Cross Validation feature selection scheme**
  - Fix home and away win percentage as base features (most indicative of the outcome)
  - Add remaining features one-at-a-time with cross-validation
  - Keep top 20 lowest-CV-error features as search set
  - Run forward selection, adding only error-reducing features, on training data
  - Stop at four features because additional inputs increase error

	Feature	Error
1	Computed Strength (A)	10.913
2	Pass yards per game allowed (H/S)	10.937
3	Points per game scored (A)	10.949
4	Total yards per game gained (H/S)	10.950
5	Computed Strength (H)	10.951
6	Total yards per game allowed (H)	10.969
7	Points per game scored (A/S)	10.979
8	Pass yards per game gained (A/S)	10.982
9	Pass yards per game gained (H/S)	11.000
10	Rush yards per game allowed (A/S)	11.006
...	...	...
39	Temperature Difference	11.161

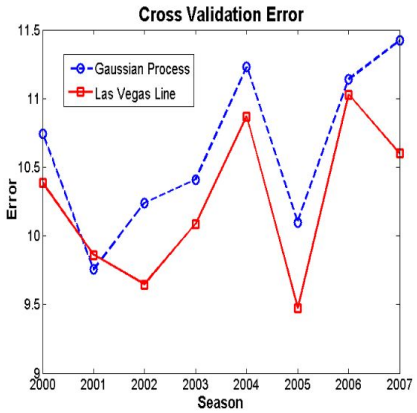
Top 10 performing features

Final Feature Set #1
Winning Percentage (H)
Winning Percentage (A)
Computed Strength (A)
Pass yards per game gained (A/S)

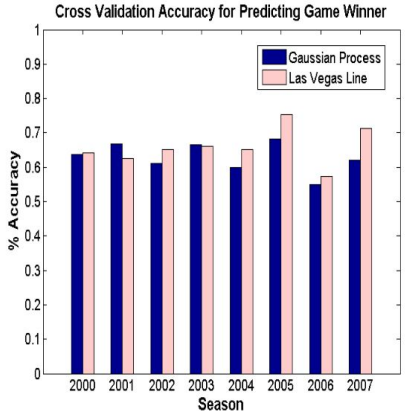
Final feature set (only four features!)

Model Implementation & Results

- **Gaussian Process Model Implementation**
  - Use a zero-mean Gaussian process with squared exponential (RBF) kernel
  - Normalize the four input features to zero mean and unit variance
  - Learn kernel hyperparameters by maximizing the log marginal likelihood
  - Output posterior mean and variance to form prediction intervals



(a)



(b)

Training data (2000-2007)

$$\frac{1}{N_{games}} \sum_{i=1}^{N_{games}} \left(|M^i_{pred} - M^i_{actual}|\right)$$

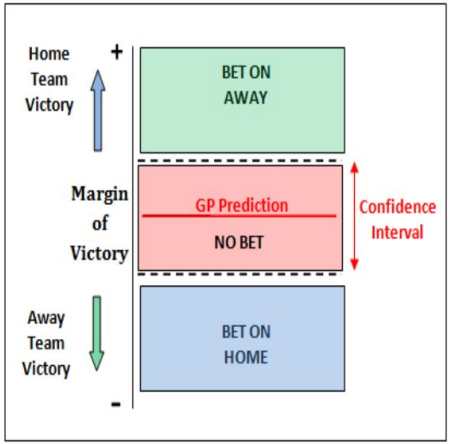
Margin of Victory Error

	Margin of Victory Error			Game Winner Accuracy		
	2008 Season	2009 Season	Mean	2008 Season	2009 Season	Mean
Gaussian Process	11.291	11.752	**11.522**	66.15%	62.56%	**64.36%**
Las Vegas Line	11.167	11.439	**11.303**	68.21%	68.72%	**68.47%**

Testing data (2008-2009)

Betting Scheme

- Compare Vegas spread to GP 95% confidence interval
- Bet home if spread below interval; bet away if above
- Place no bet when spread lies within interval
  - Notice how far less than 50% of games are bet on, another limitation of the model
- Achieved 50.9% win rate (55.2% in 2008, 48.0% in 2009), below 52.4% break-even
  - This is even with selectivity
- Vegas still wins!



Betting scheme

Year	# Games	# Bets Placed	# Bets Won	Win %
2008	195	67	37	55.22%
2009	195	98	47	47.96%
	390	165	84	50.90%

Strategy results on test data

Conclusions and Further Studies

- Vegas point spreads slightly outperform GP margin predictions: No surprise
- GP predicts game winners well (≈64%) but betting remains unprofitable
- Team strength rankings improve performance; temperature does not
- Uncertainty-based betting is principled but insufficient to beat vig

Potential Further Improvements

- Incorporate injury data and player availability into features
- Better quantify home-field advantage (travel, noise, stadium effects)
- Find a better way to incorporate more information